

Rank-two block projections contain block-rank-one projections

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Claimed status: partial result, subject to human review. The theorem fully answers the rank-two block-projection question, but not Question 6.1 for arbitrary Boolean families of projections of rank greater than one.

1 Source questions

Braga, Farah, and Vignati ask in Question 6.1 whether a sequence of projections $p_n \in C_u^*(X)$ whose every strong subseries belongs to $C_u^*(X)$ admits rank-one subprojections with the same property [1, Question 6.1, p. 20]. The exact source passage is reproduced below. The rank-two block case

6. ARE (I) AND (II) EQUIVALENT?

As mentioned earlier, the question whether the properties (I) and (II) of $C_u^*(X)$ discussed in the introduction are equivalent reduces to an innocent-looking question.

Question 6.1. Let X be a u.l.f. metric space. Suppose that p_n is a sequence of projections in $C_u^*(X)$ such that $\text{SOT-}\sum_{n \in M} p_n$ is in $C_u^*(X)$ for every $M \subseteq \mathbb{N}$ and each p_n is of rank strictly greater than 1.

1. Can we conclude that for every n there are projections $q_n \leq p_n$ of rank 1 such that $\text{SOT-}\sum_{n \in M} q_n$ is in $C_u^*(X)$ for every $M \subseteq \mathbb{N}$?
2. Can we at least conclude that there are an infinite $M \subseteq \mathbb{N}$ and $q_n \leq p_n$ of rank 1 for all $n \in M$ such that $\text{SOT-}\sum_{n \in M_0} q_n$ is in $C_u^*(X)$ for every $M_0 \subseteq M$?

A positive answer to either part of Question [6.1](#) would imply that (I) and (II) are equivalent. A partial result is given in [\[BFV21, Lemma 4.3\]](#), where a positive answer to [\(2\)](#) is given in case the projections of interest satisfy our usual regularity condition, that is, if $\inf_{n \in \mathbb{N}} \sup_{x \in X} \|p_n \chi_x\| > 0$.

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Figure 1: Question 6.1 in the source paper.

was restated as an explicit open question in Vignati’s 2026 habilitation [2, Question 5.31, p. 53]: if $p = \text{SOT-}\sum_n p_n$ is a noncompact block ghost projection and every p_n has rank two, must there be rank-one $q_n \leq p_n$ such that $\text{SOT-}\sum_n q_n \in C_u^*(X)$?

Indeed, *bad* families of projections (i.e., those giving rise to ghosts) are exactly the cause for the failure of ONL, as the following show.

Theorem 5.29. *Let (X, d) be a u.l.f. metric space. The following are equivalent:*

- *There are no noncompact block ghost projections in $C_u^*(X)$.*
- *X has ONL for equi-approximable finite-rank projections.*

Theorem 5.30. *Let (X, d) be a u.l.f. metric space. The following are equivalent:*

- *There are no noncompact block-rank-one ghost projections in $C_u^*(X)$.*
- *X has ONL for equi-approximable rank-one projections.*

It is unknown whether the existence of a noncompact block ghost projection implies the existence of a noncompact block-rank-one ghost projection. A version of the following was asked a relatively long time ago, in [14].

Question 5.31. Let X be a u.l.f. metric space, and suppose that $p = \sum p_n$ is a noncompact block ghost projection in $C_u^*(X)$ where each p_n is of rank two. Are there rank one projections $q_n \leq p_n$ such that $q = \sum q_n$ is in $C_u^*(X)$?

Having a (positive) answer to Question 5.31 would significantly enhance our understanding of ghost projections, and consequently the K -theory of uniform Roe and quasi-local algebras.

Figure 2: The rank-two block question in the 2026 habilitation.

2 Definitions and scope

Let X be a uniformly locally finite metric space. A projection $p \in C_u^*(X)$ is a *block projection* if there are pairwise disjoint finite sets $X_n \subseteq X$ such that

$$p = \text{SOT-}\sum_{n \in \mathbb{N}} p_n, \quad p_n = \chi_{X_n} p \chi_{X_n}.$$

The p_n are mutually orthogonal finite-rank projections. It is *block-rank-one* when every nonzero p_n has rank one. An operator $a \in C_u^*(X)$ is a *ghost* if its matrix coefficients tend to zero at infinity. We use only two elementary facts: diagonal multipliers χ_A lie in $\ell_\infty(X) \subseteq C_u^*(X)$, and $C_u^*(X)$ is closed under continuous functional calculus.

3 Proof intuition

Inside a two-dimensional subspace, some coordinate cut always distinguishes two orthogonal directions by a fixed amount. More precisely, for a suitable subset $A_n \subseteq X_n$, the two eigenvalues of $p_n \chi_{A_n} p_n$ are separated by at least $1/4$. There are then only finitely many spectral thresholds needed to separate the larger eigenvalue from the smaller one in every block. For each threshold class, continuous functional calculus extracts the larger eigenspace simultaneously. Adding the finitely many classes produces the desired block-rank-one projection.

4 The diagonal-gap lemma

Lemma 1 (uniform two-dimensional coordinate gap). *Let E be a two-dimensional subspace of $\ell_2(F)$, where F is finite, and let P be the orthogonal projection onto E . There is $A \subseteq F$ such that the two eigenvalues of $(P \chi_A P)|_E$ differ by at least $1/4$.*

Proof. Choose an orthonormal basis u, v of E . For $x \in F$, put

$$d_x = |u(x)|^2 - |v(x)|^2, \quad z_x = \overline{u(x)}v(x).$$

Then $\sum_x d_x = 0$ and $\sum_x z_x = 0$. For any $A \subseteq F$, the eigenvalue gap of $(P \chi_A P)|_E$ is

$$\left(\left| \sum_{x \in A} d_x \right|^2 + 4 \left| \sum_{x \in A} z_x \right|^2 \right)^{1/2}. \quad (1)$$

Set $D = \sum_x |d_x|$. If $D \geq 1/2$, take $A = \{x : d_x \geq 0\}$. Since $\sum_x d_x = 0$, $\sum_{x \in A} d_x = D/2 \geq 1/4$, and (1) gives the result. Suppose $D < 1/2$. Writing $a_x = |u(x)|^2$ and $b_x = |v(x)|^2$, we have

$$2|z_x| = 2\sqrt{a_x b_x} \geq a_x + b_x - |a_x - b_x|.$$

Thus $L := \sum_x |z_x| > (2 - 1/2)/2 = 3/4$. Averaging over $\theta \in [0, 2\pi]$ gives a θ such that

$$\sum_x \left| \text{Re}(e^{-i\theta} z_x) \right| \geq \frac{2L}{\pi}.$$

Let $A = \{x : \text{Re}(e^{-i\theta} z_x) \geq 0\}$. Because $\sum_x z_x = 0$,

$$\left| \sum_{x \in A} z_x \right| \geq \frac{L}{\pi} > \frac{3}{4\pi}.$$

The second term in (1) makes the gap larger than $3/(2\pi) > 1/4$. □

5 Rank-two block selection

Theorem 2. *Let X be a uniformly locally finite metric space and let $p \in C_u^*(X)$ be a block projection with respect to pairwise disjoint finite sets $(X_n)_{n \in \mathbb{N}}$. Suppose every nonzero block $p_n = \chi_{X_n} p \chi_{X_n}$ has rank two. Then there are rank-one projections $q_n \leq p_n$ such that*

$$q = \text{SOT-}\sum_{n \in \mathbb{N}} q_n \in C_u^*(X).$$

In fact, for every $M \subseteq \mathbb{N}$, $q_M = \text{SOT-}\sum_{n \in M} q_n$ belongs to $C_u^(X)$.*

Proof. By Lemma 1, for every n there is $A_n \subseteq X_n$ such that the two eigenvalues

$$0 \leq \lambda_n^- \leq \lambda_n^+ \leq 1 \quad \text{of} \quad h_n = (p_n \chi_{A_n} p_n)|_{p_n \ell_2(X)}$$

satisfy $\lambda_n^+ - \lambda_n^- \geq 1/4$. Let $T = \{k/16 : 1 \leq k \leq 15\}$. For every n , choose $t(n) \in T$ so that

$$\lambda_n^- + \frac{1}{32} \leq t(n) \leq \lambda_n^+ - \frac{1}{32}. \quad (2)$$

Such a grid point exists because the interval between the two endpoints has length at least $3/16$. Put $I_t = \{n : t(n) = t\}$ and $A = \bigcup_n A_n$. For $I \subseteq \mathbb{N}$, set

$$p_I = \text{SOT-}\sum_{n \in I} p_n = \chi_{\bigcup_{n \in I} X_n} p \chi_{\bigcup_{n \in I} X_n}.$$

Hence $p_I \in C_u^*(X)$. For every $t \in T$,

$$a_t = p_{I_t} \chi_A p_{I_t} = \text{SOT-}\sum_{n \in I_t} p_n \chi_{A_n} p_n$$

also belongs to $C_u^*(X)$. Choose a continuous function $f_t : [0, 1] \rightarrow [0, 1]$ that is zero on $[0, t - 1/64]$ and one on $[t + 1/64, 1]$. The margins in (2) imply that

$$q^{(t)} := f_t(a_t)$$

is the block projection whose block in $p_n \ell_2(X)$, for $n \in I_t$, is the spectral projection of h_n associated with λ_n^+ . Thus every nonzero block of $q^{(t)}$ has rank one, and continuous functional calculus gives $q^{(t)} \in C_u^*(X)$. Since T is finite,

$$q = \sum_{t \in T} q^{(t)} \in C_u^*(X).$$

Writing $q_n = \chi_{X_n} q \chi_{X_n}$ yields rank-one $q_n \leq p_n$ and $q = \text{SOT-}\sum_n q_n$. Finally, for every $M \subseteq \mathbb{N}$,

$$q_M = \chi_{\bigcup_{n \in M} X_n} q \chi_{\bigcup_{n \in M} X_n} \in C_u^*(X).$$

□

Corollary 3. *Every noncompact block ghost projection in $C_u^*(X)$ whose blocks have rank two contains a noncompact block-rank-one ghost projection. Consequently, Question 5.31 of [2] has an affirmative answer.*

Proof. Apply Theorem 2. The resulting q has infinite rank and is therefore noncompact. Since $0 \leq q \leq p$, for every $x, y \in X$,

$$|\langle q \delta_y, \delta_x \rangle|^2 \leq \langle q \delta_x, \delta_x \rangle \langle q \delta_y, \delta_y \rangle \leq \langle p \delta_x, \delta_x \rangle \langle p \delta_y, \delta_y \rangle.$$

The diagonal of the ghost projection p vanishes at infinity, so the display shows that q is a ghost. □

6 Verification, limitations, and novelty

The script `code/verify_rank_two_gap.py` numerically enumerates all coordinate subsets for random finite rank-two subspaces and checks the $1/4$ gap bound. This is only a consistency check; Lemma 1 is the proof. The result does not settle Question 6.1 for arbitrary families, nor does it show that every higher-rank block ghost projection contains a rank-two block ghost subprojection. It removes the complete rank-two block obstruction. A bounded search on 22 July 2026 covered arXiv, the published EMS version, exact phrases from Question 6.1, combinations of uniform Roe, rank two, and rank-one subprojection, and Vignati’s January 2026 habilitation. No prior proof of this rank-two selection statement was found; the habilitation still labels it Question 5.31. This is not an exhaustive novelty claim. Human review should focus on the finite-threshold functional calculus step and the ghost domination argument.

References

- [1] B. M. Braga, I. Farah, and A. Vignati, *Operator norm localization property for equi-approximable families of projections*, J. Noncommut. Geom. 18 (2024), no. 2, 657–680; arXiv:2211.02775, DOI: 10.4171/JNCG/555.
- [2] A. Vignati, *Rigidity of Roe-like algebras*, Mémoire d’habilitation à diriger des recherches, 2026, Question 5.31, p. 53.